

ARYAV TYAGI

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SUMMARY

An undergraduate student at UC San Diego passionate about engineering and innovation. I am a researcher, developer and active member of my community with experience in a multitude of high level projects deploying intuitive solutions.

EXPERIENCE

FIRST Robotics Competition

Team 3501

 08/2022 - 04/2025  Sunnyvale, United States

- Lead the first-ever vision team consisting of four people in the software subteam.
- Coded and integrated cameras using Arduinos and PhotonVision.
- Wrote and debugged Java code to autonomously complete tasks such as navigating obstacles, placing blocks, and climbing using camera readings.
- Developed code for subsystems such as the arm, swerve drive base, and PID tuning.

Project Manager

Aspiring Scholars Directed Research Program

 01/2024 - 01/2025  Fremont, United States

- Project manager who led a team of five in researching explainable AI methods such as Layer-Wise Propagation.
- Coded a convolutional neural network in Python and added multiple Layer-Wise Relevance Propagation methods to detect COVID-19 and pneumonia in CT scans and identify differences.
- Presented the results and findings of the models.

Science Olympiad

 08/2023 - 03/2025  Sunnyvale, United States

- Placed 2nd, 8th, and 10th in regionals through building wooden towers, autonomous robot cars, and elastic devices.
- Led a team of 4 in advancing in the autonomous robot car competition.

Student Leadership- ASB

 08/2024 - 06/2025  Sunnyvale, United States

- Organized school-wide events such as Olympias, races, and school dances to increase school spirit and bring the student body together.
- Hosted pie-eating challenges, cart racing, and dodgeball Olympia with over 230 students participating.
- Coordinated supplies, volunteers, and promotion for large events.
- Learned management and strategic planning skills.

Student Tutor for AP CSA

 08/2023 - 06/2024  Sunnyvale, United States

- Assisted over 30 students in learning and applying java concepts in various projects
- Helped students learn concepts such as polymorphism, data structures, recursion

KEY PROJECTS



AI Detector of Covid-19

- Trained and Tested in python a Convolutional Neural Network on CT scans of a lung through a kaggle dataset and Pytorch
- Added LRP methods such as LRP Z plus and Deep Taylor Decomposition to create heat maps that show the output decision into contributions of individual input features to see how the neural network makes decisions and how to improve it.
- Created confusion matrixes of 2 classes (normal and pneumonia) and 3 classes (normal, Covid, pneumonia) along with respective validation accuracy graphs yielding a 96.2% accuracy of the 2 classes and 97.5% of the 3 classes



FRC Robots from team 3501

- Used Organepi5, calibrating multiple arducam ov9281s along with photonVision, java code and WPI packages to detect big QR codes on the competition field and get a position on the field relative to the robot
- Using position calculated through the camera, I wrote and debugged code to autonomously drive to a certain position, use mechanisms such as saw arms, claws, elevator to complete tasks like placing blocks, tubes, and shooting rubber rings at angles
- Wrote code for arms, elevators, swerve drive along with PID tuning for systems
- Placed 2nd and 3rd at regionals, along with advancing to worlds



Autonomous robot car

- I built a 30x30 cm car powered by an arduino and stepper motors that navigates a random maze with obstacles such as walls through having the ability of changing acceleration and pivoting.



Ball Tracker

- Coded in java a NxN convolution kernel applying filters such as black and white, sharpening, blur, edge detection to track multiple colored balls on a live camera stream.

SKILLS

Java-programming language

Python-programming language

Data Structures and Algorithms

Image processing

Neural Networks

Problem Solving

Pytorch